

Artificial Intelligence and Modern Technology

Introduction

Artificial Intelligence (AI) is one of the most important areas of modern computer science. AI systems are designed to perform tasks such as understanding language, recognizing patterns, making predictions, and solving complex problems.

Machine Learning

Machine Learning (ML) is a subset of Artificial Intelligence in which computers learn patterns from data. Common types of machine learning include supervised learning, unsupervised learning, and reinforcement learning.

Natural Language Processing

Natural Language Processing (NLP) allows computers to work with human language. Applications of NLP include sentiment analysis, text classification, machine translation, question answering, chatbots, and text summarization.

Large Language Models

Large Language Models (LLMs) are neural network models trained on large collections of text. They can generate text, answer questions, summarize documents, translate languages, and assist with programming tasks.

LangChain

LangChain is a framework for developing applications powered by language models. It provides components for loading documents, splitting text, creating prompts, connecting models, retrieving information, and building AI agents.

Document Loading

Document loaders are used to bring external data into a LangChain application. For example, TextLoader can load text files, while PyPDFLoader can load PDF documents. After loading, the resulting Document objects can be passed to text splitters, embedding models, vector stores, and retrieval systems.

RAG Applications

Retrieval-Augmented Generation (RAG) combines document retrieval with language generation. A typical RAG pipeline loads documents, divides them into chunks, creates embeddings, stores the embeddings in a vector database, retrieves relevant chunks for a user's question, and sends the retrieved information to a language model.

Practice Questions

1. What is Artificial Intelligence?

2. What is the difference between AI and Machine Learning?
3. What is Natural Language Processing?
4. What is an LLM?
5. What is the purpose of a document loader in LangChain?
6. What is PyPDFLoader used for?
7. What are the main steps in a RAG pipeline?